

The Impact of Social Information on Subjective Value Representations

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Abstract (A. Schlender):

Our opinions are our own—or are they? While a broad body of research demonstrates that regions such as medial orbitofrontal cortex (mOFC) and ventromedial prefrontal cortex (vmPFC) automatically assign and encode subjective value, less is understood about how these internally generated valuations are shaped by social context. While we do generate our own opinions on things—at least initially—our evaluations are subject to change when information of what our peers believe is presented. Converging evidence suggests that social influence does more than alter outward responses; it can drive the internalization of others' beliefs by updating value representations within core valuation circuitry. Drawing on reinforcement learning frameworks, social information may function as a form of prediction error signal, modifying prior evaluations in response to discrepancies between individual judgments and perceived group norms. Additionally, because the vmPFC encodes value in a domain-general “common currency”, such socially influenced updates may generalize across both similar and dissimilar stimulus types. The present study examines how varying levels of social influence modulates individuals' subjective valuation of visual stimuli.

Introduction (A. Schlender):

We are constantly assigning value to the world around us. From trivial daily choices to more complex evaluations of people and experiences, subjective valuation is a continuous and often automatic process. Notably, this process does not always require directed cognition¹. When assessing outcome values, goal values, or decision values, a large amount of neuronal activity can be observed within vmPFC/mOFC² Table 1. Further evidence suggests that the brain encodes value even in the absence of explicit evaluation intent, with activity in the mOFC and vmPFC reliably predicting later preference choices^{3,4}. These regions are consistently implicated in representing outcome values, goals, and decision values, forming a core network for subjective valuation.

Importantly, valuation processes are not confined to deliberate decision-making contexts. Studies have shown that vmPFC/mOFC activity tracks subjective value during both attentionally demanding tasks and passive exposure, indicating that the brain continuously generates value representations⁵. Even when individuals are not instructed to evaluate stimuli, neural activity within these regions can predict subsequent preferences, suggesting that subjective valuation operates as an ongoing, implicit process.

However, humans are not isolated evaluators. As inherently social organisms, our preferences are shaped within a broader social context. A substantial body of research demonstrates that exposure to group norms—such as average attractiveness ratings—can shift individuals' subsequent evaluations. Participants who initially rate a stimulus and are then presented with a purported group consensus often adjust their ratings in the direction of that consensus. Traditionally, such effects have been interpreted as conformity driven by social pressures, including a desire to align with group norms and avoid social deviation [6, 7](#).

Crucially, emerging evidence suggests that conformity extends beyond superficial response adjustment. Social influence can lead to the internalization of others' beliefs, such that externally provided information becomes integrated into an individual's own valuation system. Neuroimaging findings show that shifts in preference following social feedback are accompanied by changes in activity within valuation-related regions, particularly the vmPFC, indicating that these effects reflect genuine updates in subjective value rather than mere compliance [8](#).

One mechanism underlying this process may be prediction error signaling. Within reinforcement learning frameworks, prediction errors arise when there is a discrepancy between expected and observed outcomes, driving the updating of value representations. In social contexts, disagreement between an individual's initial judgment and perceived group consensus may generate a "social prediction error," recruiting neural systems associated with reward learning. The magnitude of this discrepancy appears to predict the degree of subsequent preference change, implying that social information actively reshapes valuation through iterative updating processes [9, 10](#).

At the same time, neuroeconomic research indicates that the vmPFC encodes value in a domain-general manner, enabling comparisons across qualitatively different stimuli using a shared neural scale. This "common currency" framework suggests that socially derived information can be readily integrated into existing valuation processes regardless of stimulus type, allowing social influence to generalize beyond specific judgments to broader patterns of evaluation [11](#). This common currency framework implies that socially derived information—such as group preferences—can be integrated seamlessly into existing valuation processes, regardless of stimulus type. As a result, social influence has the capacity to generalize across domains, affecting not only specific judgments (e.g., attractiveness) but broader evaluative tendencies.

Taken together, these findings suggest that subjective valuation is both internally generated and dynamically shaped by social input. While initial value representations may arise automatically, these representations remain flexible and responsive to social information. The present study builds on this framework by examining how varying levels of social influence modulate individuals' subjective valuation of visual stimuli, with particular attention to whether these effects can be reliably replicated to induce conformity in individuals' internal valuation of entirely subjective preferences.

We hypothesize that social influence in the form of group feedback will be correlated with greater changes between the initial and final rapid value judgments of objects.. Furthermore, we hypothesize that the groups that receive social influence in the form of numerical and/or visual feedback of group belief, will result in the greater average difference between the pleasantness ratings than the group that does not receive any form of group feedback. We believe that the group receiving both forms of feedback will have the greatest difference between initial and final value ratings, compared to each of the groups that will only receive one form of feedback (either numerical or visual), although, each of those groups will have a greater difference between initial and final value ratings than the group that does not receive any feedback.

Methods (Primary: A. McGinty; Data Analysis written by K. Burroughs):

Link to Task: <https://ammcginty.github.io/Image-rating-task/>

Experimental Design:

The study utilizes a remote task available through a GitHub web link. Our independent variable we are manipulating between subjects is the form of feedback presented to the participant. Our dependent variable we measure is change in pre and post ratings of the same image relative to the feedback provided, if any. The study adopts four conditions, and participants are exposed to one of the following:

Condition 1: Participants are presented with a generated group distribution following first rating

Condition 2: Participants are presented with a generated group average following first rating

Condition 3: Participants are presented with both a generated group distribution and average following first rating

Condition 4: Participants are not presented with any form of feedback

Neutral images are selected to minimize the likelihood of eliciting strong emotions which may bias changes in subjective value. Individual feedback is coded in the task to generate a distribution and/or average that conflicts with each image rating. For example, if a participant's rating is between one and five, the feedback presented will be a higher average between six and then rounded to one decimal place or a distribution with values clustered around the higher end. This aims to maximize the influence of social influence on subjective value generation. Feedback is generated based on this criteria and unique to each image and for each participant. A delay period in between phase one and phase two is not included in this experiment to reduce distractibility as participants completed the task on their own time in different contexts. A primary motivation for a delay period, to reduce participant fatigue, was not a major concern as this task takes only about 3-4 minutes to complete¹³.

Participants:

Participants are friends and family members to the researchers, 18 years of age and older. A wide range of ages is selected to capture a sample representative of the general population. Participants have the opportunity to enter a given name or simply select "Start Task" in which the individual is randomly assigned a Participant ID such as "ID_5MPLO3". Thus, participants are able to perform the task with protected anonymity

if desired. When advertising the study, the researchers shared the link with an open invitation to complete the task if they so choose. Participation is entirely voluntary and no compensation, course credit, or other type of reward is delivered upon completion.

Stimuli:

Figure 1. After viewing each image, participants are provided with an opportunity to rate the image based on a scale of pleasantness.

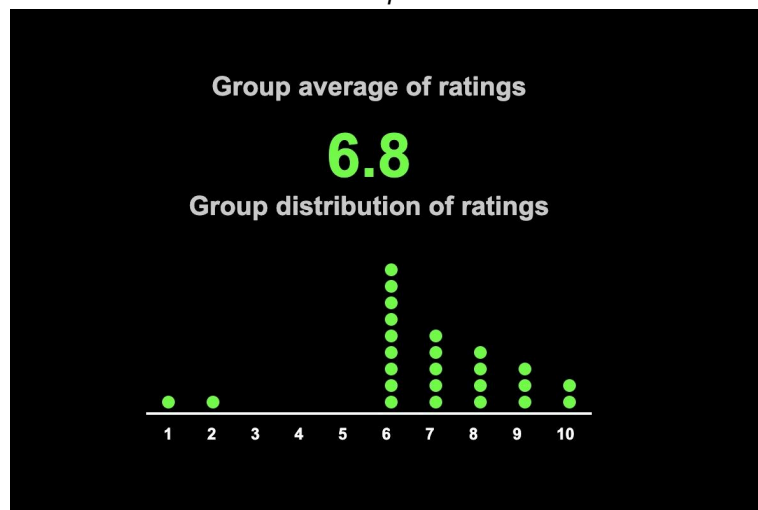


Figure 2. After providing a pleasantness rating, participants in conditions 1, 2, and 3 are presented with feedback. Note: This is a figure taken from an individual completing the task under condition 3.

Procedure:

Participants open the GitHub link and can enter a participant ID if they so choose, or may simply click “Start Task.” In phase one, participants viewed 20 neutral images for exactly 2 seconds in a set order (Image1, Image2, Image 3, etc.) then provided a rating of pleasantness on a scale of 1 (very unpleasant) to 10 (very pleasant). Individual feedback for each image was then shown, if any. In phase two, participants viewed the identical 20 images again in a randomized order and provided their second and final ratings of the images without any feedback. Thus, each participant viewed each image

twice. This order is randomized to account for order effects¹². After completing the task, the screen reads “task complete” and “data saved” then thanks the individual for their participation.

Data Analysis:

The skeleton of the task involves a database in Google Sheets, an API through Google Apps Script, creating the task through an HTML file with java script, and finally deployment through GitHub Pages. This structure allows for the collection and storage of data for further analysis. Data is then transferred to an excel file so as to be easily interpreted and utilized with the RStudio application. We have a sample size of 63 participants. Responses were screened to ensure no outliers (same rating selected for all 40 images) were included in data analyses. This comfortably places us with $n > 30$, satisfying the general guidelines of the Central Limit Theorem, while also aiming for even higher participation.

All statistical tests were conducted utilizing RStudio. The first test run is a mixed-design analysis of variance (mixed-ANOVA). The R application ensures participant-level mean ratings are calculated for initial and final ratings across trials prior to the statistical analyses. The mixed-ANOVA is then utilized to examine whether pleasantness ratings changed as a function of condition. The following statistical test assesses whether magnitude of change differed across conditions; The one-way ANOVA does this flawlessly. Finally, a linear regression analysis is conducted for the conditions with which a numerical average was shown (conditions 2 and 3), to examine whether the provided feedback resulted in adjustments of ratings per participant. The predictor variable represents the difference between the average shown and the participants initial rating, whereas the outcome variable represents change in rating from initial to final. This test allows a systemic overview of whether or not participants' ratings adjust towards the averages provided.

Results & Discussion (K. Burroughs and K. Walsh):

Results:

The mixed-ANOVA, examining effects of both condition and time on pleasantness ratings, uses condition as a between-subjects factor, and time (initial rating versus final rating) as a within-subjects factor. There is no significant effect of the condition, $F(3,59)=1.25$, $p=0.300$, $\eta^2=.060$. The effect of time also proves to be insignificant, $F(1,59)=2.86$, $p=.096$, $\eta^2=0.046$.

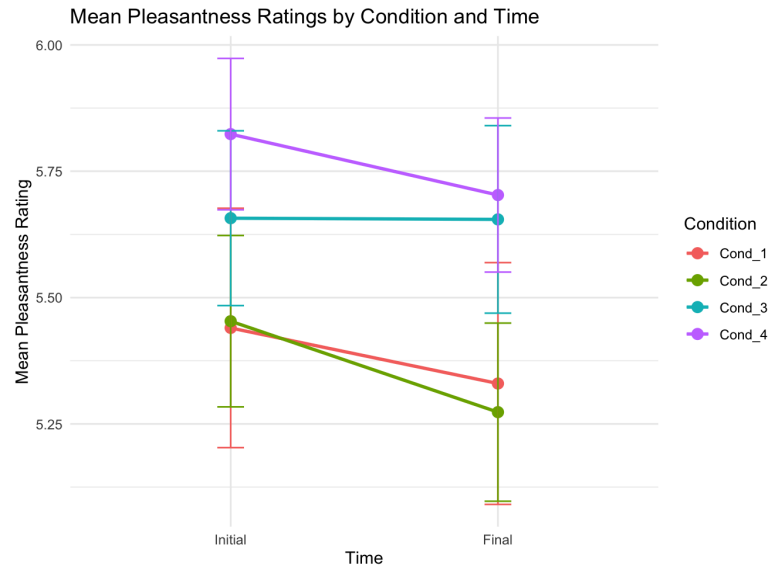


Figure 3. Mixed-ANOVA results demonstrate initial and final mean pleasantness ratings across conditions. ± 1 standard error indicated by error bars. Ratings remain stable across time and there is no significant difference in pattern of change between conditions, indicating a non-significant interaction between time and condition.

Both F-Statistics seen in the mixed-ANOVA suggest that any between group or within group differences are no greater than differences due to random noise. Both P-Values are higher than .05, indicating that our results are not statistically significant, thus we fail to reject the null hypothesis. This means that ratings did not differ between conditions, nor did the magnitude of change in ratings differ between conditions. The small effect sizes ($\eta^2 < .06$), also indicate the condition and time explain minimal variance for any changes in rating.

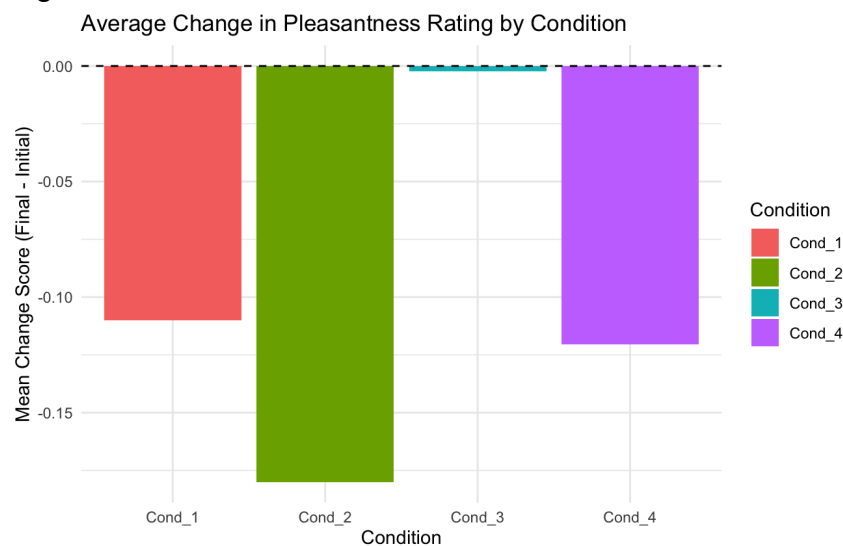


Figure 4. The one-way ANOVA demonstrates the mean change (final-initial) in pleasantness ratings per condition. Error bars represent ± 1 standard error. Small differences appear between conditions which are statistically insignificant.

The one-way ANOVA conducted on change scores reveals no significant effect of condition, $F(3,59)=0.46$, $p=.711$, $\eta^2=.023$, which indicates participants across conditions changed ratings similarly. The F-statistic further confirms what the mixed-ANOVA shows, which is that any difference between conditions may be attributed to random noise. The p-value also further confirms the results of the mixed-ANOVA, that results are statistically insignificant, thus we still can not reject the null hypothesis. The linear regression analysis reveals significant results, $F(1,718)=26.4$, $p<.001$, $R^2=.035$. The predictor variable is also a significant positive predictor of change in ratings, $\beta=0.079$, $t(718)=5.14$, $p<.001$. This indicates participants do tend to adjust their ratings in the direction of the average provided.

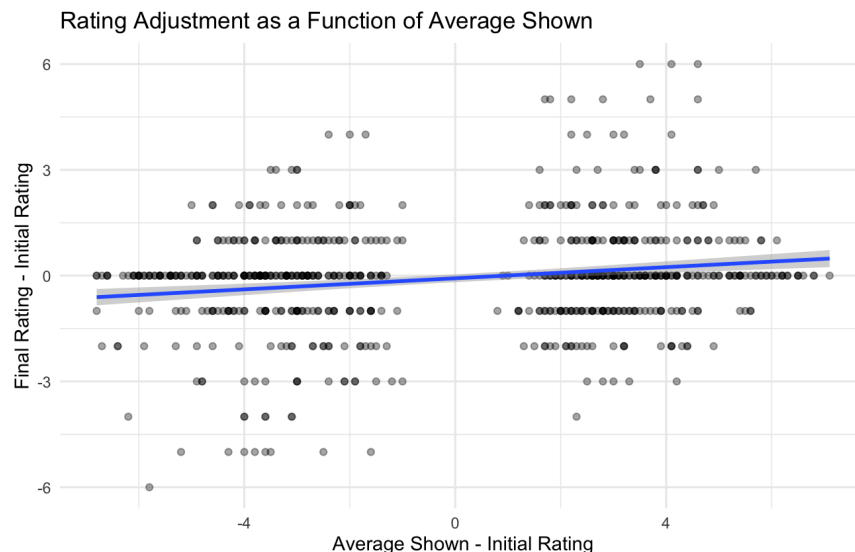


Figure 5. The regression model demonstrates the discrepancy between participant's rating adjustments and the average shown. Positive regression slope indicates significant adjustment of ratings toward average shown, given the Positive Direction values indicate the average shown was higher than participants initial ratings.

Though, the R^2 value does indicate only 3.5% of variance is explained by the average shown to participants, suggesting low-predictive power despite strong statistical significance as indicated by the F-statistic and small p-value. β informs us of the positive relationship between average shown and rating adjustments, and the high t-statistic demonstrates a low probability of results due to chance.

Discussion:

In examining the relationship between social feedback and the influence of participants' subjective value judgments we've been comparing changes in pleasantness ratings between initial image presentation and final repeated image presentation across the four conditions of feedback. We initially anticipated to observe a change in the pleasantness ratings of participants who received social feedback or stimulus than those who did not, we expected the strongest effect occurring in the combined feedback condition. The statistically insignificant results of both the mixed and one-way ANOVAs suggest that there is no significant condition-level conformity effect as we predicted. Meaning, receiving a histogram (qualitative), a class average (quantitative), both, or no

stimulus, does not significantly alter how participants rated the images the second time they see them.

The statistically significant results of the linear regression analysis suggests that participants may respond to the class average (numerical stimulus) on an individual basis. This is significant but the effect size is very small, meaning the regression is statistically significant but the practical impact of the effect was small. Overall, our findings suggest that the influence of social feedback did not significantly affect the pleasantness ratings at the group-condition level, but numerical feedback may have a small influence on individual rating adjustments.

Our results are inconsistent with what we anticipated. Given that group feedback has previously worked as a social factor pushing conformity^{6,7}, it's important for us to consider potential limitations of our study that may have impacted our results. One limitation of our study involves the feedback given to participants during the study- it may not have felt socially meaningful enough to influence participants to adjust their scores. The participants were shown a "class average" or histogram, but they might have not felt or believed that the feedback represented a real or important social group. Also, by design the stimulus intentionally displays averages opposing initial ratings the participants gave the images, i.e. if a participant rated an image a four they would be presented with an average of 6-10 and vice versa. We think this also impacted participants considering that the average were accurate because participants may have started to notice the pattern of averages disagreeing with their ratings. If the feedback did not feel credible or personally relevant, participants may have chosen to ignore it. For future research we'd attempt to combat these potential issues by ensuring that not all feedback directly opposes the participant, so as to emulate a more realistic and natural feeling source of social feedback. Additionally, the images shown in the task were stock images, which may not create a strong personal investment that would make social influence as relevant. However, as previously discussed, subjective valuation is an automatic process¹, hence we did not anticipate the type of image would affect value assignment. Nonetheless, social influence may be stronger for more personally meaningful stimuli, such as faces, moral judgements, a social opinion or a preference of the participants or the consumers, which is something we'd like to hone in on more in future research.

While the preliminary research we've been examining is an interesting start, we're particularly interested in focusing on collecting neural data in future studies. More specifically, we'd like to utilize fmri to examine current neural pathways when participants complete our current study. We'd be interested to

See how the activity in the vmPFC shifts during our task. Given that it's previously been found that social feedback was correlated with activity changes in the vmPFC as well as altered value assignments⁸, we'd be curious to see if there are any differences in how the vmPFC activates during our task, which may have caused our study to not show such prominent changes in value assignment. Additionally, we'd be interested in utilizing optogenetics to activate neurons in both the vmPFC as well as the mOFC, which are both brain regions involved in prediction of preference choice^{3,4}, to see how these two regions compare to one another in using the social feedback for valuation.

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